

Verification Plan Report

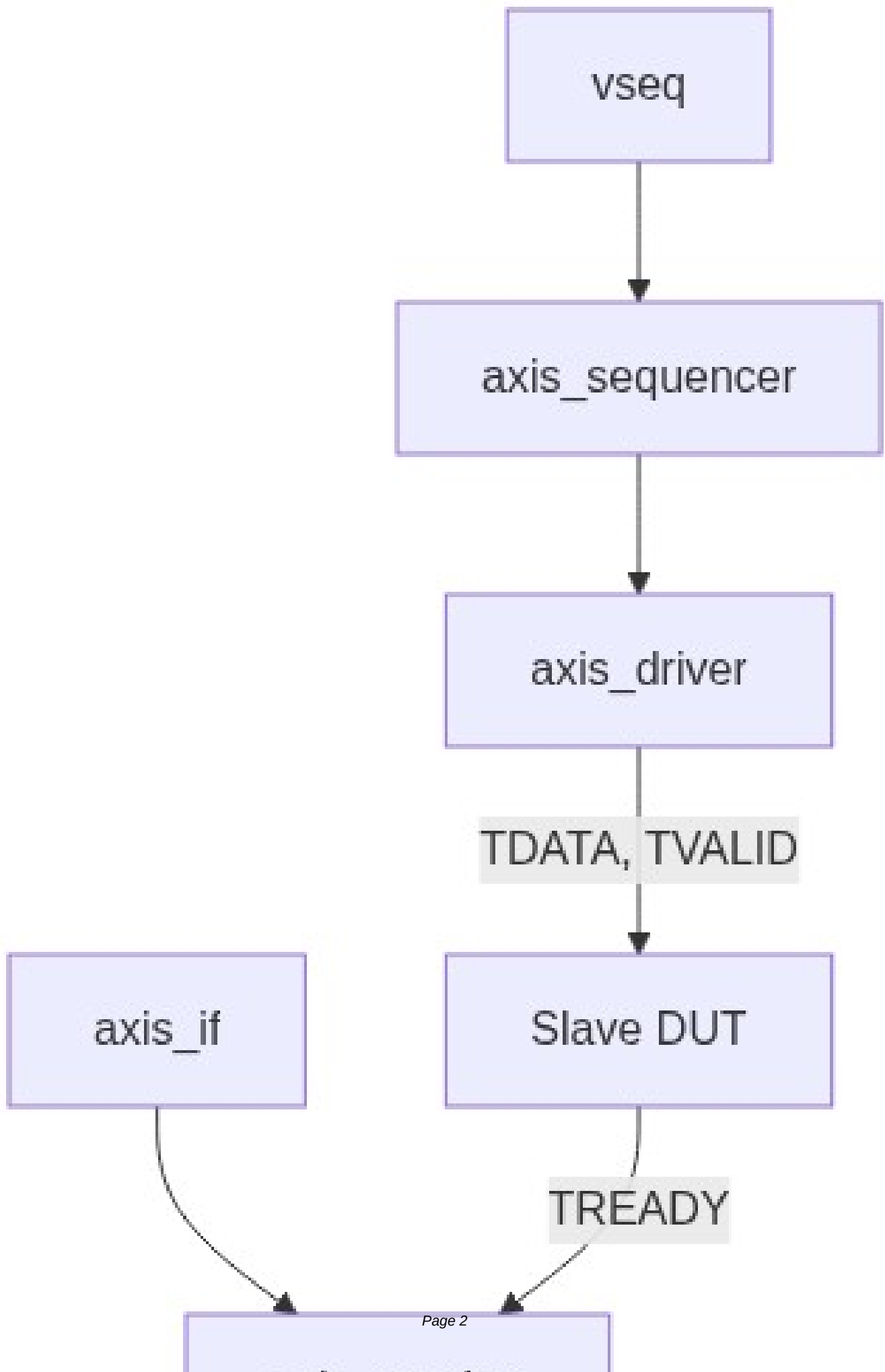
Project: AXI-Stream Enterprise Final

Version: 2.0

Verification Role: Master

1. UVM Environment Architecture

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2. Test Requirements

Index	Feature/Sub	Name	Description
REQ_01	Handshake (Stability)	Data Stability	Master must not change TDATA until TREADY is HIGH.
REQ_02	Handshake (Timing)	Valid Independence	TVALID must not depend on TREADY.
REQ_03	Reset (Timing)	Reset Suppression	TVALID must be LOW during ARESETn.
REQ_04	Reset (Recovery)	Reset Recovery	Min 1 cycle gap after ARESETn deassertion before TVALID.
REQ_05	Packet (Stability)	ID Integrity	TID/TDEST must be constant within a packet.
REQ_06	Qualifiers (TKEEP)	Null Tracking	TKEEP bits must be preserved through the stream.
REQ_07	Qualifiers (TSTRB)	Strobe Tracking	TSTRB must follow data vs position rules.
REQ_08	Packet (Boundary)	TLAST Rule	TLAST must indicate the final beat of a packet.
REQ_09	Power (AXI5)	TWAKEUP Assert	TWAKEUP must indicate activity associated with the link.
REQ_10	Sideband (TUSER)	User Sideband	TUSER must follow standard stability rules.

3. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_HS_01	Simple Handshake: Single beat transfer.	REQ_01, REQ_02
TC_HS_02	Slave Stall: Hold TREADY LOW for 20 cycles. Verify TDATA stable.	REQ_01
TC_HS_03	Master Stall: Hold TVALID LOW while TREADY 1. Verify no movement.	REQ_02
TC_HS_04	Back-to-Back: Mastery stream of 1000 continuous beats.	REQ_01
TC_RST_01	Reset Mid-Transfer: Assert reset while TVALID=1. TREADY=0.	REQ_03
TC_RST_02	Cold Boot: Drive valid on Cycle 0 of reset end. Fail expected.	REQ_04
TC_PKT_01	Single Beat Packet: TLAST=1 for 1-beat transfer.	REQ_08
TC_PKT_02	Exhaustive Packet Length: Multi-beat from 2 to 512 beats.	REQ_08
TC_PKT_03	TID Consistency: Constant TID for a 32-beat packet.	REQ_05
TC_PKT_04	TDEST Consistency: Constant TDEST for a 32-beat packet.	REQ_05
TC_PKT_05	Illegal TID Change: Toggle TID bits at Beat 5 of 10. Fail expected.	REQ_05
TC_PKT_06	Zero-Data Packet: TLAST=1 with all TKEEP bits=0.	REQ_06

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TC_QUAL_01	All-Zero TSTRB: Position-only bytes transfer.	REQ_07
TC_QUAL_02	Partial TKEEP: Sparse data stream with holes.	REQ_06
TC_QUAL_03	TSTRB Override: TKEEP=1, TSTRB=0 pattern.	REQ_07
TC_PWR_01	Wakeup Latency: TWAKEUP asserted 5 cycles before TVALID.	REQ_09
TC_PWR_02	Wakeup Drop: TWAKEUP dropped before handshake occurs. Fail expected.	REQ_09
TC_SIDE_01	TUSER Stability: Toggle TUSER while stalled. Verify violation.	REQ_10
TC_SIDE_02	TUSER Bitwalk: Walk 1 through TUSER width.	REQ_10
TC_STR_01	Random Handshake Pressure: Random TVALID/TREADY gaps.	REQ_01, REQ_02
TC_STR_02	Heavy Packet Interleaving: Rapid TID/TDEST switches.	REQ_05
TC_STR_03	Clock Frequency stress: Verify protocol at 1GHz.	REQ_01
TC_VAL_01	Minimal TVALID: Pulse TVALID only every 10 cycles.	REQ_02
TC_VAL_02	Indefinite Stall: Hold TREADY LOW for 10,000 cycles. Check stability.	REQ_01
TC_COR_01	MAX Width TDATA: 1024-bit data path throughput.	REQ_01
TC_COR_02	Zero Width TDATA: Control-only interface.	REQ_02
TC_COR_03	Max TID: Use 64-bit TID values.	REQ_05
TC_COR_04	Max TDEST: Use 64-bit TDEST values.	REQ_05
TC_FINAL_01	Regression: Sequence of 100 random packets.	REQ_01, REQ_05, REQ_08
TC_FINAL_02	Status Check: Post-regression register dump.	REQ_01
TC_EXT_01	Extra Pressure: Randomize Slave READY behavior.	REQ_01
TC_EXT_02	Extra Density: Cross TID x TLAST timings.	REQ_05
TC_EXT_03	AXI5 Wakeup Stress: Randomize power state transitions.	REQ_09
TC_EXT_04	Sideband Consistency: Verify TUSER across packet boundaries.	REQ_10
TC_EXT_05	Reset Resilience: Async reset during heavy traffic.	REQ_03

4. Functional Coverage Groups

Group Name	Description
cg_handshake	Cross of TVALID and TREADY.
cg_data	Walk-1 pattern on TDATA.
cg_packet	Packet size bins and ID switches.

5. Interface Signals

Signal Name	Direction	Width	Description
TDATA	Master	TD_W	Data payload.
TVALID	Master	1	Valid indicator.

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TREADY	Slave	1	Ready back-pressure.
TLAST	Master	1	Packet end.
TID	Master	TID_W	Stream ID.
TDEST	Master	TDEST_W	Dest ID.
TKEEP	Master	TD_W/8	Null byte marking.
TSTRB	Master	TD_W/8	Data/Pos strobe.
TUSER	Master	TU_W	User sideband.
TWAKEUP	Master	1	AXI5 Wakeup.

6. Verification Environment Structure

verif/axis_v2_0/

```
|-- agent/
|   |-- axis_driver.sv
|   |-- axis_monitor.sv
|   |-- axis_agent.sv
|   |-- axis_if.sv
|-- env/
|   |-- axis_env.sv
|   |-- axis_scoreboard.sv
|   |-- axis_pkg.sv
|-- tests/
|   |-- base_test.sv
|   |-- stress_test.sv
```

7. SystemVerilog Implementation

Master Monitor: Data Capture

```
if (vif.TVALID && vif.TREADY) begin
    axis_tx tr = axis_tx::type_id::create("tr");
    tr.data = vif.TDATA;
    tr.last = vif.TLAST;
    analysis_port.write(tr);
end
```

Master Protocol Assertion: Validity

```
property p_valid_after_reset;
    @(posedge ACLK) $rose(ARESETn) |-> !TVALID [*1];
endproperty
assert_valid_reset: assert property(p_valid_after_reset);
```